Research Experiment Results

Hierarchical Federated Learning for Privacy-Aware, Low-Latency Multimodal IoT Systems (PP-HFM)
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Complete record of all experiments executed

1. System-Level Objective Evaluation

These results correspond to the main engineering objectives of the proposed hierarchical federated multimodal system. The wearable accuracy result is treated as inconclusive where its confidence interval crosses the target threshold.

Experiment / Objective	Target	Clinical	Wearable	Status
Uplink reduction	≥ 50%	90.82%	96.08%	Pass
Round latency improvement	> 0%	16.93%	21.68%	Pass
Composed privacy budget	≤ 1	0.970	0.966	Pass
Accuracy loss	≤ 2 pp	−0.92 pp	1.97 pp	Clinical pass; wearable inconclusive
P95 inference latency	< 100 ms	4.58 ms	1.22 ms	Pass
Successful updates	> 99%	100.00%	99.94%	Pass
Energy saving	≥ 20%	72.30%	76.84%	Pass

Interpretation: seven of seven objectives pass on the clinical corpus, and six of seven on the wearable corpus with one inconclusive. The composed privacy budget includes both training and the conformal calibration release, which is stricter than the common practice of reporting the training term alone.

2. Simulation Infrastructure

Network and compute behaviour were measured on running simulators rather than derived analytically. Every network quantity reported in this document comes from rounds in which the payload actually traversed the simulated topology.

Simulator	Configuration	Measured result	Conclusion
NS-3 v3.46.1	9 IoT devices, 2 edge servers, 1 cloud; 54 Mbps access, 1 Gbps backhaul at 5 ms	Round latency 90.5 → 75.2 ms (clinical); 104.5 → 81.8 ms (wearable)	The edge tier reduces round latency by 16.93% and 21.68%
CloudSim Plus v8.5.6	IoT 300 MIPS / 1–3 W; edge 8000 MIPS / 50–200 W; GPU overhead on edge only	Centralised baseline 1266.25 J per round	Energy saving 72.30% and 76.84% against a 20% target

Interpretation: the energy figure quoted throughout is the CloudSim power-model result. A second figure of 93.5% exists from a linear bytes-to-Joules proxy, but under that model it is arithmetically identical to communication reduction and is therefore not an independent measurement of energy.

3. Baseline Comparison

Standard federated and centralised methods trained from scratch, establishing where the proposed system sits and what conventional differential privacy costs at the same budget.

Baseline	Test accuracy	Test macro-AUC	Interpretation
B0 Centralized	0.5787	0.8211	Pooled-data reference
B1 Local-Only	0.4532	0.7511	Lower reference, no collaboration
B2 FedAvg	0.6543	0.8795	Best non-private baseline
B3 FedProx	0.6570	0.8747	No advantage over FedAvg here
B4 DP-FedAvg, ε ≈ 1	0.2159	0.6443	Collapses; the like-for-like DP reference
B5 MOON	0.3934	0.7418	Underperforms on this task

Interpretation: B4 is the critical row. Applying differential privacy the conventional way at the same budget destroys the model, which is what makes the head-only result in Section 4 meaningful. These baselines are not a fair comparator for the ε = 1 head-only figure, because they train from scratch while the proposed runs start from pretrained frozen encoders; the only valid comparator is the non-private arm of the same architecture.

4. Head-Only Differentially Private Fine-Tuning

End-to-end DP-SGD at ε = 1 produced a model at chance level. The resolution was to pretrain both encoders non-privately, freeze them, and apply DP-SGD to the fusion head alone.

Recipe	Noise scale σ	Test AUC	Round time
End-to-end DP-SGD	11.3281	0.562	1046 s
Head-only DP fine-tuning	11.3281	0.8742 ± 0.0072	50 s

Interpretation: the noise scale is identical in both rows, so noise was never the limiting quantity. What changed is the optimisation geometry — a near-linear probe over frozen pretrained features tolerates σ ≈ 11.3, while training from scratch does not. Rounds also run approximately twenty times faster. The privatised head is 560,645 of 2,157,766 parameters, or 26.0%.

5. Privacy Budget Sweep

Utility across budgets under the head-only recipe, swept downward as well as upward to locate the actual utility floor.

ε	σ	AUROC	AUPRC
0.5	5.92	0.8287	0.5110
1	3.24	0.8319	0.5464
2	1.88	0.8323	0.5381
4	1.21	0.8356	0.5590
8	0.85	0.8342	0.5544
∞ (no DP)	—	0.8158	0.5204

Interpretation: utility is flat from ε = 0.5 to ε = 8, and every private arm exceeds the non-private arm. On this task differential privacy is acting as a regulariser rather than as a tax. The utility floor lies below ε = 0.5, which is lower than the budget the system operates at.

6. Accuracy Cost at ε = 1

Paired multi-seed comparison of the private arm against a non-private comparator of the same architecture, with pre-registered seed counts.

Dataset	Seeds	Accuracy cost	95% confidence interval	Verdict
VitalDB (clinical)	5	−0.92 pp	[−2.82, +0.99]	Pass
WEAR (wearable)	15	1.967 pp	[1.760, 2.173]	Inconclusive

Interpretation: on the clinical corpus the point estimate is slightly negative, so the defensible claim is no measurable cost bounded by the interval, not zero cost. On the wearable corpus the interval straddles the 2 pp threshold, so the objective can be neither passed nor failed. The seed count was pre-registered at fifteen and deliberately not extended, since extending it after observing the result would constitute p-hacking.

7. Multimodal Fusion Experiment

The multimodal result is treated as a system-setting result rather than the central scientific claim. Fusion is beneficial on VitalDB but provides little additional headroom on WEAR because the stronger unimodal model is already near ceiling.

Dataset	Best unimodal AUROC	Fusion AUROC	Fusion gain	Interpretation
VitalDB	0.8283	0.8653	+0.0370	Meaningful fusion gain
WEAR	0.9976	0.9980	+0.0004	Ceiling: fusion adds little

Interpretation: on VitalDB the gain is 5.9 times the seed standard deviation over five seeds. WEAR cannot corroborate the claim, and this is reported as a finding rather than omitted: on some IoT tasks a single modality suffices, so multimodal federated learning is not universally motivated. A correction is on record — at two seeds the gain read +0.0411 AUROC and +0.1062 AUPRC; at five seeds it is +0.0370 and +0.0671, a 37% fall in AUPRC. Only the five-seed figures should be quoted.

8. Privacy-Unit / k-Scaling Experiment

This is the main scientific experiment. Here k denotes the number of records contributed per patient. The observed record-level versus patient-level utility gap increases after k = 1, while the patient arm remains approximately flat.

k (records/patient)	Record/sample arm AUPRC	Patient arm AUPRC	Gap	p-value
1	0.5015	0.5067	−0.0052	0.75
2	0.5590	0.5084	+0.0507	0.0017
4	0.5924	0.5092	+0.0832	0.0015
8	0.5955	0.5089	+0.0866	0.0049

Interpretation: k = 1 provides a built-in null control, because at one record per patient the record and the patient are the same privacy unit and the two arms must agree. For k > 1 the record-level arm rises while the patient-level arm stays approximately flat, so the widening gap reflects the record arm improving rather than the patient arm degrading. Extra records per patient buy the patient arm nothing, because the noise scales with each patient's total contribution. This supports an observed records-per-patient dependence; external validation is still required before calling it a universal law.

Reporting limits: AUROC follows the same monotone shape (+0.0006, +0.0094, +0.0133, +0.0132) but no value of k reaches significance at three seeds, so AUROC is reported as directionally consistent and not as a resolved effect. Accuracy is pinned at 0.8005 in every patient row, which reflects an uncalibrated 0.5 threshold on imbalanced data, and should not be quoted.

9. Patient-Level versus Record-Level Differential Privacy

The cost of stating the guarantee over patients rather than over training records.

Quantity	Measured value	Interpretation
Corpus structure	48,712 windows across 6,257 cases; mean 7.79 windows per case	Every patient contributes more than one record
Patients contributing > 1 record	100%	The two guarantees differ for every individual in the dataset
Cost of the unit upgrade	≈ 1.77 × σ	Stronger privacy unit has a measurable optimisation cost

Interpretation: because not one patient in the corpus contributes only a single record, a record-level guarantee and a patient-level guarantee are different statements for every individual present. This is the premise on which the project rests, and it is measured rather than assumed.

10. Membership-Inference Audit

The original MIA hypothesis predicted a substantially larger patient-level attack advantage for record-level DP. The paired LiRA results did not establish that large effect. A positive control demonstrated that the attack can detect leakage where leakage exists.

Arm	MIA AUC	Original interpretation
Non-private control	0.6277	Positive control: attack signal present
Record-level DP, ε ≈ 1	0.5477	Small signal; large predicted effect not established
Patient-level DP, ε ≈ 1	0.5224	Near chance; large predicted effect not established

Attack variant	Paired Δ (record − patient)	95% confidence interval	p-value
Loss threshold	+0.0138	[−0.0034, +0.0310]	0.090
LiRA offline	+0.0325	[−0.0280, +0.0929]	0.210
LiRA online	+0.0034	[−0.0059, +0.0128]	0.365
LiRA online, global variance	−0.0211	[−0.0487, +0.0065]	0.101

Interpretation: every confidence interval includes zero. The predicted effect of approximately 0.20 lies far outside all four intervals and is refuted. A small effect is not excluded, since the half-width on the strongest attack is ±0.060 at five seeds. The correct wording is that the large predicted effect is refuted while a small one remains unresolved at this sample size. A supporting diagnostic is that on the patient arm LiRA adds nothing over a plain loss threshold (−0.0071), which is the signature of a defence with no difficulty-confounded signal left rather than of an underpowered adversary.

11. MIA Difficulty-Floor Control

A new control corrects an important evaluation issue: members and non-members differ in intrinsic difficulty. A shadow-only attacker without the target model establishes a difficulty floor. Attack performance should therefore be interpreted relative to this floor rather than only against AUC = 0.5.

Arm	Attack AUC	Difficulty floor	Margin over floor
Positive control	0.6254	0.5626	+0.0627
Record DP, ε ≈ 1	0.5477	0.5252	+0.0226
Patient DP, ε ≈ 1	0.5224	0.5167	+0.0057
Record DP, ε ≈ 16	0.5465	0.5743	−0.0278
Patient DP, ε ≈ 16	0.5630	0.5259	+0.0371

Interpretation: the positive control still clears its difficulty floor, supporting the validity of the attack. The ε ≈ 1 private arms clear their floors only slightly, and the ε ≈ 16 record arm falls below its floor. This makes the refutation of the originally predicted large MIA effect stronger. Members are drawn from the released training patients and non-members from the released test patients, and those pools differ in intrinsic difficulty (mean statistic 10.25 against 7.71), so an attack is otherwise rewarded for detecting difficulty rather than membership. Floors are computed per arm from that arm's own shadow models, which is why they differ; better-fitting models separate easy from hard patients more sharply.

12. Membership Inference at a Loosened Budget

A final test of whether the membership hypothesis failed only because ε = 1 left no headroom for a gap to appear. At ε = 16 the noise scale is thirteen times lower (σ 11.33 falling to 0.85), so a membership signal should survive if one exists.

Attack variant	Paired Δ (record − patient)	95% confidence interval	Reading
Loss threshold	+0.0163	[+0.0123, +0.0204]	Significant, predicted direction, but an eighth of the predicted size
LiRA offline	−0.1788	[−0.2553, −0.1024]	Significant in the reversed direction; an artefact of over-correction
LiRA online	+0.0057	[−0.0137, +0.0252]	Chance
LiRA online, global variance	−0.0552	[−0.0896, −0.0208]	Significant in the reversed direction

Interpretation: the four attacks contradict one another. Two report a significant gap in the direction opposite to the prediction, one reports a significant gap an eighth of the predicted size, and the principled two-sided variant reports chance. Mutually inconsistent estimators are the signature of no underlying signal, with each estimator driven by a different facet of the difficulty confound. Decisively, the record-level arm sits 0.0278 below its own difficulty floor, so it carries no membership signal for a gap to be composed of; loosening the budget sixteenfold changed its attack AUC by 0.0012. The hypothesis that the effect was merely hidden at ε = 1 is therefore refuted, and the question is closed rather than left open at small sample size.

Note for reporting: the LiRA offline value of −0.1788 must be presented together with its explanation. It arises because the record arm's offline score is 0.3842, far below chance, produced by over-correction on an asymmetric difficulty tail. Presented without that explanation it would be misread as evidence that patient-level privacy leaks more.

13. Privacy Budget and Training Configuration

Configuration	Key result	Interpretation
Patient-level matched-budget DP	ε ≈ 0.97 composed	Meets the ε ≤ 1 target
Training + conformal calibration	0.7705 + 0.2000 = 0.9705	Composition remains below 1
Patient-level upgrade	≈ 1.77 × σ against record-level	Stronger privacy unit has a measurable optimisation cost
Head-only DP	AUC 0.8742; much faster, stronger utility	Parameter-efficient private adaptation is effective
Rigorous public pretraining	−0.022 AUROC against oracle pretraining	The cost of a guarantee that is valid end-to-end

14. Ablation Experiments

Each arm removes one component from the full system, three seeds per arm on the clinical corpus.

Removed component	AUROC	Uplink	Conclusion
None (full system)	0.8541 ± 0.0035	251.97 MB	Reference configuration
Hierarchy	0.8484 ± 0.0057	50.40 MB	Hierarchy mainly supports systems efficiency
Privacy	0.8307 ± 0.0060	251.97 MB	Removing DP makes the model worse, not better
Sparsification	0.8553 ± 0.0015	1259.75 MB	+0.0012 AUROC for five times the traffic
Quantisation	0.8543 ± 0.0036	403.14 MB	+0.0002 AUROC for 1.6 times the traffic
Multimodality	0.8283	—	Fusion contributes; see Section 7
Added ASWA aggregation	0.8542 ± 0.0037	251.97 MB	+0.0005 AUROC; no measurable contribution

Interpretation: three results follow. Removing differential privacy makes the model worse, confirming the sweep in Section 5 — DP is regularising rather than taxing on this task. Both compression stages earn their place, since dense transmission buys +0.0012 AUROC for five times the traffic and float32 buys +0.0002 for 1.6 times. ASWA contributes nothing and has been deleted from the codebase; it was a leftover from a scrapped earlier design.

These arms had never produced a result until four silent defects were fixed on 16 August 2026: a dead command-line argument rejected by the parser, a quantiser with a hardcoded 8-bit path, a summary reading test columns from a file containing only validation columns, and a verdict threshold set at 1e-4.

15. Communication and Compression

Measurement	Bytes per round	Interpretation
Full-model FP32 FedAvg	258,931,920	Reference
Head-only, uncompressed	67,277,400	Effect of freezing the encoders
Observed, after defect fix	16,821,300	90.82% reduction against the reference

Interpretation: a defect was found and corrected during this work. The compressor took a fixed fraction of every tensor, so frozen encoders, whose delta is exactly zero, still contributed 20% of their elements to each payload as explicit zeros. Omitting all-zero tensors accounted for a 78.87% reduction on its own. The fix was proven not to alter learning: the dense path is bit-identical to the previous implementation over five rounds, and an isolated re-run moved AUC by 0.0001.

16. Inference Latency and Edge Deployment

Precision	Provider	Model size	Total P95	Status
FP32	CPU	10.64 MB	16.363 ms	Pass; recommended
FP32	GPU (CUDA)	10.64 MB	22.476 ms	Pass
INT8	CPU	6.55 MB	31.028 ms	Pass but slower than FP32
INT8	GPU (CUDA)	6.55 MB	73.150 ms	Pass but substantially slower

Interpretation: the privacy apparatus is free at serving time. The conformal set construction that carries the coverage guarantee occupies 0.68% of the inference path, and the private mechanism occupies none of it, because its noise is applied to gradients during training. The objection that a privacy apparatus is too heavy for a latency-bound deployment is therefore answered by measurement rather than by argument.

INT8 quantisation is a characterised negative result. It is 1.90 times slower on CPU and 3.25 times slower on GPU than FP32, and misses the 4 MB size target at 6.55 MB. Dynamic quantisation accelerates only matrix-multiply operations, while this model's cost is dominated by convolutions and a recurrent path; and on CUDA the quantised operations have no GPU kernel and fall back to CPU, with the runtime reporting 3180 memory-copy nodes inserted into the graph. The FP32 artifact is the one to deploy.

17. Feature Attribution

Integrated gradients over released checkpoints, twenty runs across five recipes and two baselines, with a separate seed-stability check.

Recipe	Share of attribution carried by missingness masks
Full-model DP	28.2%
Patient-level DP	22.0%
Record-level DP	21.6%
Full-model, no DP	19.9%
Head-only, no DP	15.8%

Interpretation: differential privacy shifts the model's attention toward missingness. Under noise the model relies more on which measurements exist than on their values, and the shift is monotone in how much privacy is applied. Rank correlation remains 0.95 between the head-only private and non-private recipes. For the related claim that missingness carries signal, masks account for 22.0% of attribution mass over five seeds and one observed indicator reaches 4.99% ± 0.33, ranking in the top five on four of five seeds; this should be described as consistently among the strongest rather than assigned a fixed rank, and requires a mask ablation before it constitutes proof.

18. Negative and Corrected Results

Recorded in full, because the credibility of the positive results depends on them.

Item	Outcome
Predicted patient-level MIA vulnerability	Refuted at ε = 1 and again at ε = 16, with a passing positive control on both occasions
Adaptive per-group privacy budget (SADP)	Underperforms uniform DP at equal ε by construction; the cross-group composition tax drives σ from 11.3 to 70
INT8 edge quantisation	Slower than FP32 on both providers and misses the size target
ASWA aggregation	+0.0005 AUROC; deleted from the codebase
Fusion gain at two seeds	Corrected downward at five seeds; AUPRC fell 37%
Legacy corpus	Withdrawn entirely; its modalities were paired by a construction rule rather than at source, so its accuracy results were not evidence of fusion
Wearable accuracy objective	Inconclusive; seed count pre-registered and deliberately not extended
Wearable fusion	At ceiling; structurally unable to corroborate the multimodal claim
Head-only DP fine-tuning	Memorises members harder than full-model training, 0.6277 against 0.5122; a caveat to our own recipe

19. Experiments Not Yet Run

Experiment	Why it matters
Window-count-normalised patient clipping	The only remaining candidate for an original mechanism rather than a measurement. A patient's gradient is summed over k windows and then clipped to a fixed bound, so long recordings are systematically down-weighted; normalising before clipping should correct this.
External clinical validation on eICU-CRD	Tests whether the records-per-patient dependence in Section 8 is structural or specific to one corpus. Awaiting PhysioNet credentialed access. The prediction must be frozen before the data are seen.
Conformal coverage validation	The coverage guarantee is implemented and its budget cost is composed into the reported ε, but empirical coverage has not been measured.
Mask-channel ablation	Would convert the attribution evidence in Section 17 into proof.
Hierarchical arm without DP	Completes the topology axis alongside the existing ablation arms.
Multi-seed baselines	Would place confidence intervals on the baseline comparisons in Section 3.

20. Summary of Outcomes

Category	Count	Note
Experiments completed	24	All finished; none abandoned mid-run
Succeeded outright	14	Including all engineering objectives
Characterised negative results	5	Reported rather than dropped
Mixed or inconclusive	3	Wearable accuracy, wearable fusion, attribution evidence
Superseded	1	The end-to-end budget sweep, retained only as a comparison arm
Cost priced rather than passed or failed	1	Rigorous public pretraining

Overall: the engineering objectives are met with wide margins and are not in doubt. The central scientific finding is the records-per-patient dependence in Section 8, which carries its own null control at k = 1. The originally predicted membership-inference vulnerability was tested twice under a validated attack and refuted on both occasions, and is retained as a reported negative result. External validation on an independent clinical corpus is the principal outstanding item.